

7.4 Classwork: Solving Exponential Equations — Markscheme

1. [Maximum mark: 8]

(a) $3^x = 17$. Take log of both sides: $x \log 3 = \log 17$. **M1**

$$x = \frac{\log 17}{\log 3} = 2.5789 \dots = 2.58 \text{ (3 s.f.)} \quad \textbf{A1}$$

(b) $4^{x-2} = 30$. $(x-2) \log 4 = \log 30$. **M1**

$$x-2 = \frac{\log 30}{\log 4} = 2.4534 \dots, \text{ so } x = 4.4534 \dots = 4.45 \text{ (3 s.f.)} \quad \textbf{A1}$$

(c) $2^{3x+1} = 100$. $(3x+1) \log 2 = \log 100$. **M1**

$$3x+1 = \frac{\log 100}{\log 2} = \frac{2}{\log 2} = 6.6438 \dots, \text{ so } x = \frac{6.6438 \dots - 1}{3} = 1.8813 \dots = 1.88 \text{ (3 s.f.)} \quad \textbf{A1}$$

(d) $5^x = 7^{x-1}$. Take log: $x \log 5 = (x-1) \log 7$. Collect x -terms: $x(\log 7 - \log 5) = \log 7$. **M1**

$$x = \frac{\log 7}{\log 7 - \log 5} = \frac{0.8451}{0.8451 - 0.6990} = 5.7853 \dots = 5.79 \text{ (3 s.f.)} \quad \textbf{A1}$$

2. [Maximum mark: 7]

$$P(t) = 200 \cdot 3^{t/10}.$$

(a) At $t = 0$: $P(0) = 200 \cdot 3^0 = 200.$ **A1**

(b) $P(24) = 200 \cdot 3^{24/10} = 200 \cdot 3^{2.4}.$ **M1**

$= 200 \cdot 13.9666 \dots = 2793.3 \dots \approx 2793$ **A1**

(c) $200 \cdot 3^{t/10} = 5000 \Rightarrow 3^{t/10} = 25$ **M1 A1**

Take log: $\frac{t}{10} \log 3 = \log 25$, so $\frac{t}{10} = \frac{\log 25}{\log 3} = 2.9299 \dots$ **M1**

$t = 29.299 \dots \approx 29.3$ hours (1 d.p.) **A1**

3. [Maximum mark: 7]

$$M(t) = 50 \cdot \left(\frac{1}{2}\right)^{t/8}.$$

(a) $M(20) = 50 \cdot (0.5)^{20/8} = 50 \cdot (0.5)^{2.5}.$ **M1**

$= 50 \cdot 0.17677 \dots = 8.8388 \dots \approx 8.84 \text{ g (3 s.f.)}$ **A1**

(b) 60 % decayed means 40 % remains, so $M(t) = 20$:

$50 \cdot (0.5)^{t/8} = 20 \Rightarrow (0.5)^{t/8} = 0.40$ **M1**

Take log: $\frac{t}{8} = \frac{\log 0.40}{\log 0.5} = \frac{-0.39794}{-0.30103} = 1.3219 \dots$ **M1**

$t = 8 \cdot 1.3219 \dots = 10.575 \dots \approx 10.6 \text{ days (3 s.f.)}$ **A1**

(c) $50 \cdot (0.5)^{t/8} = 1 \Rightarrow (0.5)^{t/8} = \frac{1}{50} = 0.02$ **M1**

$\frac{t}{8} = \frac{\log 0.02}{\log 0.5} = 5.6439 \dots, \text{ so } t = 45.15 \dots \approx 45 \text{ days}$ **A1**

4. [Maximum mark: 6]

$$T(t) = 20 + 65 \cdot (0.9)^t.$$

(a) $T(0) = 20 + 65 \cdot 1 = 85^\circ\text{C}.$ **A1**

(b) $T(15) = 20 + 65 \cdot (0.9)^{15}.$ **M1**

$= 20 + 65 \cdot 0.20589 \dots = 20 + 13.383 \dots = 33.383 \dots \approx 33.4^\circ\text{C} \text{ (3 s.f.)}$ **A1**

(c) Set $T(t) = 40$:

$$20 + 65 \cdot (0.9)^t = 40 \Rightarrow (0.9)^t = \frac{20}{65} = \frac{4}{13} \approx 0.30769$$
 M1

Take log: $t \log 0.9 = \log\left(\frac{4}{13}\right)$, so $t = \frac{\log(4/13)}{\log 0.9} = \frac{-0.51217}{-0.04576}$ **M1**

$t = 11.193 \dots \approx 11.2 \text{ minutes (3 s.f.)}$ **A1**